

## I. Client Engagement

### A. Project Overview

**StockFinder PH** is a web application that allows customers to search for products and check their stock availability in nearby partnered stores. It features an interactive map that helps customers locate stores close to their area, saving them time and travel costs by confirming product availability before visiting.

### B. Identification of Customer Requirements

- Customer Requirements:
  - Customers waste time and gas visiting stores that are out of stock.
  - Phone calls to stores go unanswered during busy hours.
  - Difficulty locating nearby stores that carry needed items.
  - No easy way to check “What’s available nearby” before shopping.
- Store Owner Requirements
  - Difficulty competing with large online retailers.
  - Limited budget for online advertising and visibility.
  - No platform to showcase inventory to potential customers.

### C. Proposed Solution:

StockFinder Ph addresses these challenges by providing a centralized platform where customers can easily search for products, view real-time stock availability, and locate nearby stores using an interactive map interface.

- Core Features
  - ✓ Product search across multiple partnered stores
  - ✓ Real-time stock availability display
  - ✓ Interactive map showing nearby locations
  - ✓ Store information including contact details and operating hours
  - ✓ Simple admin panel for store owners to update inventory

### D. Stakeholder Identification

**Primary Stakeholder - Client:** Store owners who partner with StockFinder PH to list their inventory.

**End Users:** Customers who use the platform to search for products.

**Project Team:** Development team responsible for building and deploying the system.

**System Administrator:** Manages platform operation and technical infrastructures.

### E. Requirements Gathering

#### a. Functional Requirements

1. User registration and authentication for store owners
2. Product catalog management (add,edit,delete)
3. Real-time inventory tracking and updates
4. Search functionality with filters (category, location, and availability)
5. Interactive map integration showing store locations
6. Stock status indicators (In Stock, Low Stock, and Out of Stock)
7. Store profile management
8. Mobile-responsive interface

#### b. Non-Functional Requirements

1. System response time under 2 seconds
2. Support for 100+ concurrent users
3. Mobile device compatibility
4. Browser compatibility (Chrome, Firefox)
5. Data security and privacy compliance

- 6. Uptime availability
- 7. Scalable architecture for future growth

F. Contract Signing and Project Launch  
Upon agreement of all requirements, timeline and deliverables, the project is signed.

- Project Duration: 2.5 months
- Deliverable Functional web application
- Payment Terms: Milestone-based payments
- Support Period: 3 months post-launch
- Warranty Period: 6 months for bugs fixes

II. Project Planning

A. Description of the Project:  
**StockFinder PH** is a location-based web application designed to connect customers with local stores through real-time inventory visibility. The platform enables customers to search for specific products, view current stock levels across multiple stores. StockFinder PH focuses exclusively on inventory visibility, allowing customers to verify product availability before physically visiting stores, saving time and costs.

The system benefits both customers (saving time and transportation costs) and store owners (attracting targeted foot traffic and competing with online retailers).

B. Technology to be used:

| Component   | Technology                                                      |
|-------------|-----------------------------------------------------------------|
| Frontend    | HTML, CSS and JavaScript                                        |
| Backend     | PHP                                                             |
| Database    | MySQL                                                           |
| Server      | Apache                                                          |
| Other Tools | phpMyAdmin for database management and Git for version control. |

C. Project Schedule

| Phase                     | Tasks                                           | Duration  |
|---------------------------|-------------------------------------------------|-----------|
| 1. Requirements Gathering | Documentation, interviews, scope definition     | Week 1    |
| 2. System Design          | HLD, LLD, wireframes                            | Week 2    |
| 3. Frontend Development   | UI pages, search UI                             | Week 3-4  |
| 4. Backend Development    | Database setup, API creation, stock query logic | Weeks 4-6 |
| 5. Integration            | Connecting frontend + backend + map             | Week 7    |
| 6. Testing                | Unit testing, integration,                      | Weeks 8-9 |

|               |                                 |         |
|---------------|---------------------------------|---------|
|               | system testing                  |         |
| 7. Deployment | Final deployment, user training | Week 10 |

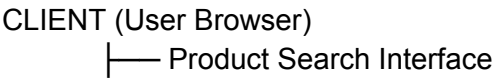
#### D. Team Composition

| Member                             | Role            | Responsibilities                                                                                                                                    |
|------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Ronde, Richie                      | Client          | Provides project requirements, clarifies business needs, reviews deliverables, and approves final outputs.                                          |
| Arnado, Norwejean                  | System Analyst  | Analyzes requirements, prepares system specifications, creates HLD/LLD alignment, and ensures the system fits business needs.                       |
| Magsayo, Owen Robert S.            | Project Manager | Oversees project planning, scheduling, resource allocation, risk management, and ensures the project is completed on time and within scope.         |
| Borne, Kier                        | Project Leader  | Leads the development team, assigns tasks, monitors progress, coordinates with the Project Manager, and ensures technical decisions are consistent. |
| Aledon, Stephen<br>Durano, Neokent | Dev             | Implements frontend and backend features, integrates APIs, performs unit testing, fixes bugs, and contributes to system documentation.              |

#### High Level Design (HLD)

Describes the software’s logical structure, information flow, logical processing steps control logic data structures.

- High-level Architecture (logical view)



- └─ Interactive Map (Stores)
- └─ Store & Product Details Page
- └─ Login/Registration UI

#### BACKEND SERVER (API + Business Logic)

- └─ Authentication Management
- └─ Product Search & Filtering
- └─ Store Location & Mapping Services
- └─ Inventory / Stock Status Handling
- └─ Notification Manager (future)

#### DATABASE (Data Storage)

- └─ User Data
- └─ Store Data (with geolocation)
- └─ Product Data
- └─ Inventory Data (Store ↔ Product link)

## 2. Information Flow

### A. Product Search

1. User enters product search keyword/location
2. Frontend sends request to Backend API
3. Backend looks up products in database
4. Backend filters stores by location and product availability
5. Backend returns results with stock status and distance
6. Frontend displays products and store markers on the map

### B. Inventory Update

1. User enters product search keyword/location
2. Frontend sends request to Backend API
3. Backend looks up products in database
4. Backend filters stores by location and product availability
5. Backend returns results with stock status and distance
6. Frontend displays products and store markers on the map

## 3. Logical Processing Steps

### A. User Registration/ Login

- authenticate and create accounts.

#### Steps:

1. The user opens the Registration/Login page.
2. The user enters details (email/phone, password, name).
3. System validates input (format, mandatory fields).
4. System checks if email/phone already exists:
  - If yes → show error.
  - If no → continue.
5. Passwords are hashed and stored in the database.
6. Generate authentication token (JWT or session).
7. Return success response to frontend.

8. User is redirected to dashboard or search page

## B. Product Search

- Allow users to find products and see nearby stores with stock.

### Steps:

1. User enters search keyword + optional filters (category, brand, location, radius).
2. Frontend sends API request
3. Backend receives request:
  - Sanitize input to prevent injection.
  - Extract location info; if missing, prompt user to enter manually.
4. Backend performs product search:
  - Full-text search on product names and descriptions.
  - Filter by category/brand if provided.
5. Backend retrieves nearby stores:
  - Geo-query: find stores within radius using
  - Only include approved stores.
6. Backend checks stock status for each product in each store:
  - Quantity > threshold → In Stock
  - Quantity ≤ threshold → Low Stock
  - Quantity = 0 → Out of Stock
  - No recent update → Unknown
7. Backend calculates distance from user to each store.
8. Backend sorts results:
  - First by distance
  - Then by availability
  - Optional: sort by price or popularity
9. Backend returns JSON response with:
  - Product info
  - Stores + stock status + distance
10. Frontend displays:
  - Product list with key info
  - Interactive map with store markers
  - Optional “sort/filter” UI
11. Users can click a store or product to see more details.

### C. Store Inventory Update

- Allow stores to update stock availability.

Steps:

1. Store logs into the dashboard.
2. Store chooses:
  - Manual update (form)
  - Upload CSV
  - POS integration (if available)
3. Backend authenticates the store.
4. Backend validates input:
  - Product IDs exist
  - Stock quantities are numeric
5. Backend updates inventory table:
  - Update quantity, stock\_status, last\_updated timestamp
6. Backend invalidates cache for affected products/stores.
7. If any customer watchlist exists for updated products:
  - Queue notification jobs (email/push/SMS)
8. Return success message to store.

### D. Watchlist / Notification Handling

- Notify users when desired products are in stock.

Steps:

1. Users add products to the watchlist via UI.
2. Frontend sends API request
3. Backend stores user\_id + product\_id + target stock status.
4. When inventory updates occur:
  - Check if updated stock matches watchlist conditions.
  - If yes → create notification record.
5. Send notification to the user (email, push, SMS).
6. Mark notification as “sent” or “read” after delivery.

## E. Map Display & Store Location Logic

- Help users locate stores visually.

Steps:

1. Frontend fetches product + store data from API.
2. For each store:
  - Get **geo\_lat** and **geo\_lng**.
  - Calculate distance from user location.
3. Display map:
  - Marker per store
  - Info window shows store name, stock status, distance
4. Optional: cluster markers for dense areas
5. Optional: allow click → link to external map/directions

## F. Control Logic

Stock Status Decision:

- if quantity > threshold → In Stock
- if quantity ≤ threshold → Low Stock
- if quantity = 0 → Out of Stock
- if last\_updated > N hours → Unknown

Store Visibility:

- Only approved stores shown
- Only stores within user radius

Search Result Sorting:

- Distance → Stock availability → Price → Popularity

User Location Handling:

- GPS available → use coordinates
- GPS denied → request manual input

## G. Error Handling / Edge Cases

- Invalid input → return error message
- No products found → return empty array + suggestion

- Store deleted or unapproved → do not include in results
- Inventory data stale → mark stock as “Unknown”

H. Optional Advanced Steps

- POS webhook integration for real-time inventory
- Product recommendations based on past searches
- Analytics for user behavior and popular stores/products
- Delivery or order placement feature

4. Data Structures

A. Entity Relationship Diagram (ERD)

- Core Database Table

**Users:** User accounts, authentication credentials, roles

**Stores:** Store information, location (lat/long), contact details, operating hours

**Products:** Product catalog with name, description and price

**Categories:** Product categorization for filtering

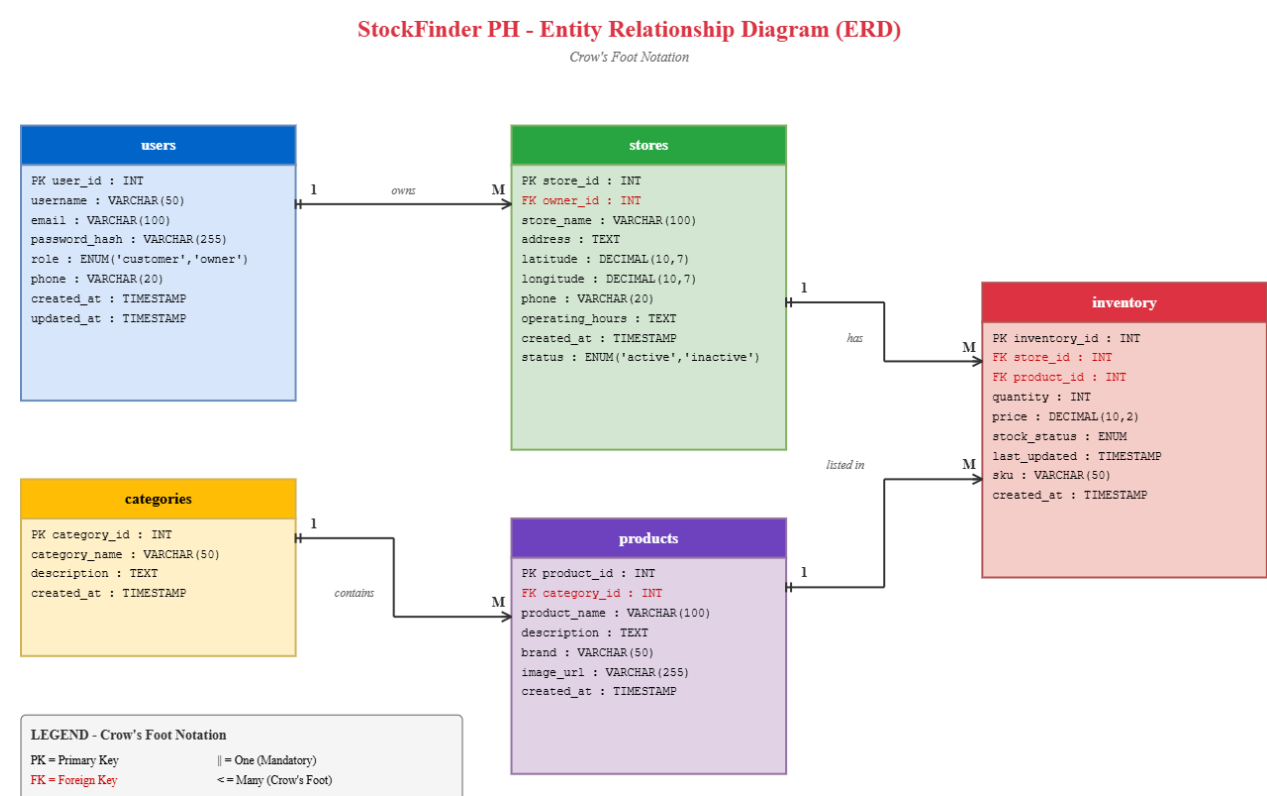


Figure 1.1 ERD

B. Table Descriptions

Table 1: users

Purpose: Store user accounts for customers and store owners.



| Column Name   | Data Type                        | Constraints                           | Description                                                                 |
|---------------|----------------------------------|---------------------------------------|-----------------------------------------------------------------------------|
| user_id       | INT                              | PRIMARY KEY, AUTO_INCREMENT, NOT NULL | Unique identifier for each user                                             |
| username      | VARCHAR(50)                      | UNIQUE, NOT NULL                      | User's login username (3-50 characters, alphanumeric)                       |
| email         | VARCHAR(100)                     | UNIQUE, NOT NULL                      | User's email address (must be valid format)                                 |
| password_hash | VARCHAR(255)                     | NOT NULL                              | Hashed password using bcrypt (cost factor 10)                               |
| role          | ENUM('customer','owner','admin') | DEFAULT 'customer'                    | User role: customer (view only), owner (manage stores), admin (full access) |
| phone         | VARCHAR(20)                      | NULLABLE                              | Optional contact phone number                                               |
| created_at    | TIMESTAMP                        | DEFAULT CURRENT_TIMESTAMP             | Account creation timestamp                                                  |
| updated_at    | TIMESTAMP                        | ON UPDATE CURRENT_TIMESTAMP           | Last profile update timestamp                                               |

Table 2: stores  
Purpose: Store information with geolocation data for interactive map display.

| Column Name | Data Type     | Constraints                            | Description                                           |
|-------------|---------------|----------------------------------------|-------------------------------------------------------|
| store_id    | INT           | PRIMARY KEY, AUTO_INCREMENT            | Unique identifier for each store                      |
| owner_id    | INT           | FOREIGN KEY → users(user_id), NOT NULL | References user who owns this store (CASCADE DELETE)  |
| store_name  | VARCHAR(100)  | NOT NULL                               | Name of the store (must be unique per owner)          |
| address     | TEXT          | NOT NULL                               | Complete physical address of the store                |
| latitude    | DECIMAL(10,7) | NOT NULL                               | Geographical latitude (-90 to 90) for map plotting    |
| longitude   | DECIMAL(10,7) | NOT NULL                               | Geographical longitude (-180 to 180) for map plotting |
| phone       | VARCHAR(20)   | NOT NULL                               | Store contact phone number                            |

| Column Name     | Data Type                           | Constraints               | Description                                                                              |
|-----------------|-------------------------------------|---------------------------|------------------------------------------------------------------------------------------|
| operating_hours | TEXT                                | NULLABLE                  | Store operating hours (can be JSON or plain text)                                        |
| created_at      | TIMESTAMP                           | DEFAULT CURRENT_TIMESTAMP | Store registration timestamp                                                             |
| status          | ENUM('active','inactive','pending') | DEFAULT 'active'          | Store status: active (visible in search), inactive (hidden), pending (awaiting approval) |

Table 3: categories  
Purpose: Product categorization for filtering and organization

| Column Name   | Data Type   | Constraints                 | Description                                            |
|---------------|-------------|-----------------------------|--------------------------------------------------------|
| category_id   | INT         | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for each category                    |
| category_name | VARCHAR(50) | UNIQUE, NOT NULL            | Name of the category (e.g., 'Electronics', 'Pharmacy') |
| description   | TEXT        | NULLABLE                    | Optional description of category contents              |
| created_at    | TIMESTAMP   | DEFAULT CURRENT_TIMESTAMP   | Category creation timestamp                            |

Table 4: products  
Purpose: Master product catalog shared across all stores.

| Column Name  | Data Type    | Constraints                                     | Description                                        |
|--------------|--------------|-------------------------------------------------|----------------------------------------------------|
| product_id   | INT          | PRIMARY KEY, AUTO_INCREMENT                     | Unique identifier for each product                 |
| category_id  | INT          | FOREIGN KEY → categories(category_id), NOT NULL | References product category (RESTRICT DELETE)      |
| product_name | VARCHAR(100) | NOT NULL                                        | Name of the product (searchable, indexed)          |
| description  | TEXT         | NULLABLE                                        | Detailed product description (searchable, indexed) |

|       |             |          |                            |
|-------|-------------|----------|----------------------------|
| brand | VARCHAR(50) | NULLABLE | Product brand/manufacturer |
|-------|-------------|----------|----------------------------|

|            |              |                           |                                              |
|------------|--------------|---------------------------|----------------------------------------------|
|            |              |                           | (searchable, indexed)                        |
| image_url  | VARCHAR(255) | NULLABLE                  | URL or path to product image (must be < 5MB) |
| created_at | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP | Product creation timestamp                   |

Table 5: inventory  
Purpose: Links products to stores with stock quantity and pricing information  
(Many-to-Many relationship)

| Column Name  | Data Type                                   | Constraints                                  | Description                                                                  |
|--------------|---------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------|
| inventory_id | INT                                         | PRIMARY KEY, AUTO_INCREMENT                  | Unique identifier for each inventory record                                  |
| store_id     | INT                                         | FOREIGN KEY → stores(store_id), NOT NULL     | References the store (CASCADE DELETE)                                        |
| product_id   | INT                                         | FOREIGN KEY → products(product_id), NOT NULL | References the product (CASCADE DELETE)                                      |
| quantity     | INT                                         | NOT NULL, DEFAULT 0, CHECK (quantity >= 0)   | Current stock quantity (must be non-negative)                                |
| price        | DECIMAL(10,2)                               | NOT NULL, CHECK (price >= 0)                 | Product price at this store (2 decimal places)                               |
| stock_status | ENUM('in_stock','low_stock','out_of_stock') | GENERATED/STORED                             | Auto-calculated: out_of_stock (qty=0), low_stock (qty 1-5), in_stock (qty>5) |
| last_updated | TIMESTAMP                                   | DEFAULT CURRENT_TIMESTAMP ON UPDATE          | Automatically updated when quantity or price changes                         |
| sku          | VARCHAR(50)                                 | NULLABLE                                     | Store-specific SKU/barcode identifier                                        |
| created_at   | TIMESTAMP                                   | DEFAULT CURRENT_TIMESTAMP                    | Inventory record creation timestamp                                          |

Low Level Phase  
Test case creation  
Unit Test (UT) and Integration Test/System Test (IT/ST)

# UNIT TESTING

## 1. Login - Positive

|                       |                           |                |        |
|-----------------------|---------------------------|----------------|--------|
| Test Scenario ID      | SCN-01                    | Test Case ID   | TC-001 |
| Test Case Description | Correct login credentials | Test Priority  | High   |
| Pre-Requisite         | N/A                       | Post-Requisite | N/A    |

| S.No | Action                                          | Inputs                        | Expected Output                                                                            | Actual Output | Test Browser | Test Result |
|------|-------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------|---------------|--------------|-------------|
| 1    | Open the website and navigate to the Login Page | URL:<br>www.stockfinderph.com | The website and login page loads successfully                                              |               | Chrome       |             |
| 2    | Enter valid username                            | Username:<br>Kali             | Username accepted                                                                          |               | Chrome       |             |
| 3    | Enter valid password                            | Password:<br>password123      | Password is masked and accepted                                                            |               | Chrome       |             |
| 4    | Click the Login button                          | N/A                           | System validates credentials                                                               |               | Chrome       |             |
| 5    | Verify landing page                             | N/A                           | Customer: redirected to StockFinder PH search page<br>Owner: redirected to Owner Dashboard |               | Chrome       |             |

## 2. Login - Negative

|                       |                             |                |        |
|-----------------------|-----------------------------|----------------|--------|
| Test Scenario ID      | SCN-02                      | Test Case ID   | TC-002 |
| Test Case Description | Incorrect login credentials | Test Priority  | High   |
| Pre-Requisite         | SCN-01                      | Post-Requisite | N/A    |

| S.No | Action                                          | Inputs                        | Expected Output       | Actual Output | Test Browser | Test Result |
|------|-------------------------------------------------|-------------------------------|-----------------------|---------------|--------------|-------------|
| 1    | Open the website and navigate to the Login Page | URL:<br>www.stockfinderph.com | The website will show |               | Chrome       |             |

|   |                          |                       |              |  |        |  |
|---|--------------------------|-----------------------|--------------|--|--------|--|
| 2 | Input incorrect username | Username: Kali        | Login failed |  | Chrome |  |
| 3 | Input incorrect password | Password: password213 | Login failed |  | Chrome |  |

3. Functionality of Register Button

|                       |                                  |                |        |
|-----------------------|----------------------------------|----------------|--------|
| Test Scenario ID      | SCN-03                           | Test Case ID   | TC-003 |
| Test Case Description | Functionality of register button | Test Priority  | High   |
| Pre-Requisite         | US-01                            | Post-Requisite | N/A    |

| S.No | Action                          | Inputs                     | Expected Output        | Actual Output | Test Browser | Test Result |
|------|---------------------------------|----------------------------|------------------------|---------------|--------------|-------------|
| 1    | Open the website                | URL: www.stockfinderph.com | The website will show  |               | Chrome       |             |
| 2    | User clicks the register button | N/A                        | Register page displays |               | Chrome       |             |

4. Product Search Module - Positive (Customer)

|                       |                                                                 |                |                                         |
|-----------------------|-----------------------------------------------------------------|----------------|-----------------------------------------|
| Test Scenario ID      | SCN-04                                                          | Test Case ID   | TC-004                                  |
| Test Case Description | Verify that searching an existing product shows correct results | Test Priority  | High                                    |
| Pre-Requisite         | Product exists in database.                                     | Post-Requisite | Search results are displayed on screen. |

| S.No | Action                                              | Inputs    | Expected Output                                    | Actual Output | Test Browser | Test Result |
|------|-----------------------------------------------------|-----------|----------------------------------------------------|---------------|--------------|-------------|
| 1    | Navigate to the Search Page                         | N/A       | Search page loads properly                         |               | Chrome       |             |
| 2    | Enter product name in the search bar                | iPhone 14 | Typed value appears in input field                 |               | Chrome       |             |
| 3    | Click the Search button                             | N/A       | Product list displayed                             |               | Chrome       |             |
| 4    | Verify that the search results contain correct data | iPhone 14 | Correct product details and availability are shown |               | Chrome       |             |

5. Product Search Module - Negative (Customer)

|                       |                                                 |                |                         |
|-----------------------|-------------------------------------------------|----------------|-------------------------|
| Test Scenario ID      | SCN-05                                          | Test Case ID   | TC-005                  |
| Test Case Description | Verify system behavior for nonexistent products | Test Priority  | Medium                  |
| Pre-Requisite         | User on search page                             | Post-Requisite | Error message displayed |

| S.No | Action                               | Inputs      | Expected Output            | Actual Output               | Test Browser | Test Result |
|------|--------------------------------------|-------------|----------------------------|-----------------------------|--------------|-------------|
| 1    | Navigate to the Search Page          | N/A         | Search page loads properly | Website opened successfully | Chrome       |             |
| 2    | Enter product name in the search bar | xyz123phone | “No products found”        |                             | Chrome       |             |
| 3    | Verify map behavior                  | N/A         | No pins displayed          |                             | Chrome       |             |

6. Stock Availability Module (Customer)

|                       |                                 |                |                        |
|-----------------------|---------------------------------|----------------|------------------------|
| Test Scenario ID      | SCN-06                          | Test Case ID   | TC-006                 |
| Test Case Description | Verify stock quantity retrieval | Test Priority  | High                   |
| Pre-Requisite         | Product and stock exist         | Post-Requisite | Availability displayed |

| S.No | Action                   | Inputs     | Expected Output                    | Actual Output | Test Browser | Test Result |
|------|--------------------------|------------|------------------------------------|---------------|--------------|-------------|
| 1    | Search product           | Samsung TV | Product list displayed             |               | Chrome       |             |
| 2    | Verify stock             | N/A        | Store list appears                 |               | Chrome       |             |
| 3    | Check stock availability | N/A        | Shows “In Stock” or “Out of Stock” |               | Chrome       |             |

7. Add Product (Owner)

|                       |                                                                             |                |                                                  |
|-----------------------|-----------------------------------------------------------------------------|----------------|--------------------------------------------------|
| Test Scenario ID      | SCN-07                                                                      | Test Case ID   | TC-007                                           |
| Test Case Description | Verify that the owner can successfully add a new product with valid inputs. | Test Priority  | High                                             |
| Pre-Requisite         | Owner is logged in and on the Owner Dashboard Add/Edit Product form.        | Post-Requisite | New product appears in the “Your Products” list. |

| S.No | Action | Inputs | Expected | Actual | Test | Test Result |
|------|--------|--------|----------|--------|------|-------------|
|------|--------|--------|----------|--------|------|-------------|

|   |                                     |                |                                                           |        |         |  |
|---|-------------------------------------|----------------|-----------------------------------------------------------|--------|---------|--|
|   |                                     |                | Output                                                    | Output | Browser |  |
| 1 | Click inside the Product Name field | MacBook Pro 14 | Input appears in the field                                |        | Chrome  |  |
| 2 | Select a category from dropdown     | Laptops        | Category is selected                                      |        | Chrome  |  |
| 3 | Enter product quantity              | 10             | Quantity appears in the field                             |        | Chrome  |  |
| 4 | Click Save Product                  | N/A            | System waves product                                      |        | Chrome  |  |
| 5 | Verify product appears in listing   | N/A            | Product “MacBook Pro 14 (10pcs)” appears in Your Products |        | Chrome  |  |

8. Edit Product (Owner)

|                       |                                                                  |                |                                           |
|-----------------------|------------------------------------------------------------------|----------------|-------------------------------------------|
| Test Scenario ID      | SCN-08                                                           | Test Case ID   | TC-008                                    |
| Test Case Description | Verify that the owner can successfully edit an existing product. | Test Priority  | High                                      |
| Pre-Requisite         | Product already exists in the list.                              | Post-Requisite | Updates product appears with new details. |

| S.No | Action                      | Inputs                               | Expected Output                    | Actual Output | Test Browser | Test Result |
|------|-----------------------------|--------------------------------------|------------------------------------|---------------|--------------|-------------|
| 1    | Click Edit beside a product | N/A                                  | Product details auto-fill the form |               | Chrome       |             |
| 2    | Modify Product Name         | MacBook Air (M2) → MackBook Air (M3) | Updated name appears               |               | Chrome       |             |
| 3    | Modify Quantity             | 2 → 5                                | Updated quantity appears           |               | Chrome       |             |
| 4    | Click Save Product          | N/A                                  | Product updates in the list        |               | Chrome       |             |
| 5    | Verify listing              | N/A                                  | MacBook Air (M3) - 5 pcs appears   |               | Chrome       |             |

9. Delete Product (Owner)

|                       |                                                              |                |                                   |
|-----------------------|--------------------------------------------------------------|----------------|-----------------------------------|
| Test Scenario ID      | SCN-09                                                       | Test Case ID   | TC-009                            |
| Test Case Description | Verify that the owner can delete a product from the listing. | Test Priority  | High                              |
| Pre-Requisite         | Product exists in “Your Products”.                           | Post-Requisite | Product is removed from the list. |

| S.No | Action                        | Inputs       | Expected Output                   | Actual Output | Test Browser | Test Result |
|------|-------------------------------|--------------|-----------------------------------|---------------|--------------|-------------|
| 1    | Click Delete beside a product | N/A          | Confirmation prompt appears       |               | Chrome       |             |
| 2    | Confirm deletion              | Click Yes/Ok | Product is removed                |               | Chrome       |             |
| 3    | Verify listing                | N/A          | Deleted product no longer visible |               | Chrome       |             |

INTEGRATION TEST

| Test Case ID | Test Case Description                                                     | Pre-Conditions                                                                                                                                                    | Test Case                             | Test Steps                                                                                                                                                                                                                                                                                                                                                                                                            | Expected Test Results                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ITC-001      | Verify complete customer registration, login, and product search workflow | Application is accessible and running<br>- Database is empty or has no user with email "kali@test.com"<br>- At least one store with products exists in the system | Customer Registration and Log In Flow | <div>1. Navigate to login-register.html page</div> <div>2. Click on "Register" tab in Customer section</div> <div>3. Enter registration details:<div>- Username: "kali"</div><div>- Email: "kali@test.com"</div><div>- Password: "password123"</div></div> <div>4. Click "Register" button</div> <div>5. Verify success message and automatic login</div> <div>6. Verify redirection to index.html (main search</div> | <div>-Registration successful, user account created in database</div> <div>- User automatically logged in after registration</div> <div>- Redirected to search page (index.html)</div> <div>- Search returns relevant MacBook products</div> <div>- Store markers display correctly on map</div> <div>- Store popup shows name, address, phone, hours</div> <div>- Product details modal shows product name, category, quantity, price, stock status</div> <div>- Session persists across page navigation</div> |



|         |                                                                                                   |                                                                                                                                                                                     |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                   |                                                                                                                                                                                     |                                                 | <div>page)</div> <div>7. Enter product name "MacBook" in search bar</div> <div>8. Click "Search" button</div> <div>9. Verify product results display with store information</div> <div>10. Click on a store marker on the map</div> <div>11. Verify store details popup appears</div> <div>12. Click "Details" button on a product</div> <div>13. Verify product details modal opens with correct information</div>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ITC-002 | Verify store owner can register, login, and manage complete product lifecycle (add, edit, delete) | <div>- Application is accessible and running</div> <div>- Database has no user with email "owner@test.com"</div> <div>- Categories table populated with at least 3 categories</div> | Store Owner Registration and Product Management | <div>1. Navigate to login-register.html page</div> <div>2. Click on "Register" tab in Store Owner section</div> <div>3. Enter registration details:<div>- Username: "test_owner"</div><div>- Email: "owner@test.com"</div><div>- Password: "Owner123@"</div></div> <div>4. Select your Store</div> <div>5. Click "Register" button</div> <div>6. Verify success message and automatic login</div> <div>7. Verify redirection</div> | <div>- Store owner registration successful</div> <div>- Owner automatically logged in</div> <div>- Redirected to owner dashboard</div> <div>- Product "iPhone 15 Pro" added successfully with quantity 10</div> <div>- Product displays with green "In" badge (In Stock)</div> <div>- Product editable, quantity updated to 3</div> <div>- Stock status changes to yellow "Low" badge</div> <div>- Product editable again, quantity updated to 0</div> <div>- Stock status changes to red "Out" badge</div> <div>- Product successfully deleted</div> <div>- Product no longer visible in inventory list</div> <div>- Database records updated correctly at</div> |

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|  |  |  |  | <div>to<br/>owner-dash<br/>board.html</div> <div>8. In "Add/Edit<br/>Product"<br/>section,<br/>enter:<br/>- Product Name:<br/>"iPhone 15 Pro"<br/>- Category:<br/>Select<br/>"Electronics" from<br/>dropdown<br/>- Quantity: 10</div> <div>9. Click<br/>"Save<br/>Product"<br/>button</div> <div>10. Verify<br/>product<br/>appears in<br/>"Your<br/>Products"<br/>list with "In<br/>Stock"<br/>status</div> <div>11. Click "Edit"<br/>button on<br/>the newly<br/>added<br/>product</div> <div>12. Modify<br/>quantity to<br/>3</div> <div>13. Click "Save<br/>Product"<br/>button</div> <div>14. Verify<br/>product<br/>now shows<br/>"Low<br/>Stock"<br/>status (qty<br/>1-5)</div> <div>15. Click "Edit"<br/>button<br/>again</div> <div>16. Modify<br/>quantity to<br/>0</div> <div>17. Click "Save<br/>Product"<br/>button</div> <div>18. Verify<br/>product<br/>now shows<br/>"Out of<br/>Stock"<br/>status</div> <div>19. Click<br/>"Delete"<br/>button on<br/>the product</div> <div>20. Confirm<br/>deletion in<br/>popup</div> <div>21. Verify</div> | each step |
|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|

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|         |                                                                                     |                                                                                                                                                       |                                             | product removed from "Your Products" list                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| ITC-003 | Verify customer sees real-time inventory updates when store owner modifies products | -Store owner logged in on owner-dashb oard.html<br>- Customer logged in on index.html<br>- At least one product "Dell XPS 13" exists with quantity 10 | Real-Time Inventory Update Synchronizatio n | <ol style="list-style-type: none"><li>1. [Tab 2 - Customer] Search for "Dell XPS 13"</li><li>2. Verify product appears with "In Stock" status showing "15 pcs"</li><li>3. [Tab 1 - Owner] Navigate to owner dashboard</li><li>4. Click "Edit" on "Dell XPS 13" product</li><li>5. Change quantity from 15 to 2</li><li>6. Click "Save Product"</li><li>7. [Tab 2 - Customer] Observe search results WITHOUT manual refresh</li><li>8. Verify product automatical ly updates to show "2 pcs" with "Low Stock" status</li><li>9. [Tab 1 - Owner] Click "Edit" on "Dell XPS 13" again</li><li>10. Change quantity from 2 to 0</li><li>11. Click "Save Product"</li><li>12. [Tab 2 - Customer] Observe search results</li></ol> | <ul style="list-style-type: none"><li>- Customer initially sees "Dell XPS 13" with 15 pcs, "In Stock"</li><li>- After owner updates to 2 pcs, customer page auto-refreshes within 1-2 seconds</li><li>- Customer now sees "2 pcs" with yellow "Low Stock" badge</li><li>- After owner updates to 0 pcs, customer page auto-refreshes again</li><li>- Customer now sees "Out of Stock" or product disappears from results</li><li>- No manual page refresh required</li><li>- localStorage sync works via storage event listener</li></ul> |

|         |                                                                                                                                      |                                               |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|         |                                                                                                                                      |                                               |                                                                                                                                                                                                                                                    | WITHOUT<br>manual<br>refresh<br>13. Verify product<br>automatically<br>updates to show<br>"Out of Stock"<br>status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| ITC-004 | Verify<br>product<br>search with<br>combined<br>filters<br>(category,<br>availability,<br>location)<br>returns<br>correct<br>results | Product<br>Search with<br>multiple<br>filters | - Customer<br>logged in on<br>index.html<br>- Database has<br>products in<br>multiple<br>categories<br>- Database has<br>products with<br>varying stock<br>levels (In, Low,<br>Out)<br>- At least 3<br>stores exist<br>with different<br>locations | <ol style="list-style-type: none"><li>1. Navigate to<br/>(search<br/>page)</li><li>2. Enter<br/>search term<br/>"Lenovo" in<br/>search bar</li><li>3. Select<br/>Category<br/>filter: "All<br/>categories"</li><li>4. Select<br/>Availability<br/>filter: "All"</li><li>5. Click<br/>"Search"<br/>button</li><li>6. Note the<br/>number of<br/>results<br/>returned<br/>(e.g., 5<br/>products)</li><li>7. Change<br/>Category<br/>filter to<br/>"Laptops"</li><li>8. Click<br/>"Search"<br/>button<br/>again</li><li>9. Verify<br/>results<br/>filtered to<br/>only show<br/>Lenovo<br/>laptops</li><li>10. Change<br/>Availability<br/>filter to "In<br/>Stock"</li><li>11. Click<br/>"Search"<br/>button</li><li>12. Verify<br/>results<br/>show only<br/>Lenovo<br/>laptops with<br/>quantity &gt; 5</li><li>13. Click "Use<br/>My<br/>Location"<br/>button</li><li>14. Allow<br/>location<br/>access<br/>when</li></ol> | - Initial search returns<br>all Lenovo products<br>(any category, any<br>stock)<br>- Category filter<br>reduces results to<br>only Laptops<br>- Availability filter<br>further reduces to<br>only In Stock items<br>- Stock status<br>correctly reflects: In<br>(qty > 5), Low (qty<br>1-5), Out (qty = 0)<br>- Geolocation sorts<br>stores by distance<br>(nearest first)<br>- View toggle<br>switches between<br>product list and store<br>list<br>- Map markers<br>update based on<br>filtered results<br>- Store popup shows<br>only products<br>matching current<br>filters |

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|         |                                                                                          |                                                                                                            |                             | <div>browser prompts</div> <div>15. Verify stores sorted by distance from user location</div> <div>16. Change View filter to "Stores"</div> <div>17. Verify display switches from products to store-centric view</div> <div>18. Click on nearest store marker on map</div> <div>19. Verify popup shows matching products available at that store</div>                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ITC-005 | Verify customer can save favorites, export to file, and import on another device/browser | <div>- Customer logged in on index.html</div> <div>- At least 5 different products exist in database</div> | Favorites Export and Import | <div>1. Search for product "MacBook Pro 14"</div> <div>2. Click star icon to add to favorites</div> <div>3. Search for product "Dell XPS 13"</div> <div>4. Click star icon to add to favorites</div> <div>5. Search for product "Sony WH-1000X M4"</div> <div>6. Click star icon to add to favorites</div> <div>7. Check "Only favorites" checkbox</div> <div>8. Verify only the 3 favorited products display</div> <div>9. Click "Export Favorites" button</div> <div>10. Verify</div> | <div>- Star icons fill when products added to favorites</div> <div>- "Only favorites" filter shows exactly 3 products</div> <div>- Export button downloads valid JSON file</div> <div>- JSON file contains array with 3 product objects</div> <div>- After localStorage clear, favorites are empty</div> <div>- Import successfully restores all 3 favorites</div> <div>- Favorites persist across sessions after import</div> |

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|         |                                                                                                                   |                                                                                                                                                                                                            |                                      | <div>browser<br/>downloads<br/>file named<br/>"stockfinder-<br/>favorites-[date].json"</div> <div>11. Open<br/>downloaded JSON file<br/>and verify it<br/>contains 3<br/>product<br/>entries</div> <div>12. Clear<br/>browser<br/>localStorage<br/>(simulate<br/>new<br/>device)</div> <div>13. Refresh<br/>page</div> <div>14. Verify<br/>favorites list<br/>is empty</div> <div>15. Click<br/>"Import<br/>Favorites"<br/>button</div> <div>16. Select the<br/>previously<br/>exported<br/>JSON file</div> <div>17. Click<br/>"Open"</div> <div>18. Check<br/>"Only<br/>favorites"<br/>checkbox<br/>again</div> <div>19. Verify the 3<br/>previously<br/>saved<br/>products<br/>reappear</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| ITC-006 | Verify<br>interactive<br>map displays<br>stores<br>correctly and<br>provides<br>navigation to<br>external<br>maps | <div>- Customer<br/>logged in on<br/>index.html</div> <div>- At least 3<br/>stores exist<br/>with valid<br/>latitude/longitude<br/>coordinates</div> <div>- User<br/>location<br/>access<br/>granted</div> | Map<br>Interaction and<br>Directions | <div>1. Navigate to<br/>index.html</div> <div>2. Search for<br/>any product<br/>(e.g.,<br/>"laptop")</div> <div>3. Verify map<br/>displays<br/>with<br/>multiple<br/>store<br/>markers</div> <div>4. Verify each<br/>marker<br/>color-coded<br/>by stock<br/>status:</div> <div>- Green: In Stock</div> <div>- Yellow: Low<br/>Stock</div> <div>- Red: Out of<br/>Stock</div>                                                                                                                                                                                                                                                                                                                   | <div>- Map loads with<br/>correct number of<br/>store markers</div> <div>- Markers placed at<br/>correct geographic<br/>coordinates</div> <div>- Marker colors reflect<br/>current stock status</div> <div>- Info window<br/>displays correct store<br/>information</div> <div>- "Open in Maps" link<br/>works and opens<br/>Google Maps</div> <div>- Google Maps shows<br/>correct location with<br/>marker</div> <div>- Only one info<br/>window open at a<br/>time</div> <div>- Store details modal<br/>shows</div> |

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|         |                                                                              |                                                                                                                        |                                             | <div>5. Click on first store marker</div> <div>6. Verify info window opens with:<ul style="list-style-type: none"><li>- Store name</li><li>- Product availability</li><li>- Address</li><li>- Phone number</li></ul></div> <div>7. Click "Open in Maps" link in info window</div> <div>8. Verify Google Maps opens in new tab with store location</div> <div>9. Close Google Maps tab</div> <div>10. Click different store marker on StockFinder map</div> <div>11. Verify previous info window closes and new one opens</div> <div>12. Click "Details" button in info window</div> <div>13. Verify store details modal opens showing:<ul style="list-style-type: none"><li>- Store name</li><li>- Full address</li><li>- Phone (clickable)</li><li>- Email (clickable)</li><li>- Operating hours</li><li>- List of products with stock levels</li></ul></div> | <div>comprehensive information</div> <ul style="list-style-type: none"><li>- Phone and email links are clickable</li><li>- Product list in modal matches current inventory</li></ul>             |
| ITC-007 | Verify customer can compare same product availability across multiple stores | <div>- Customer logged in on index.html</div> <div>- Product "Dell XPS 13" exists in at least 3 different stores</div> | Multi-Store Product Availability Comparison | <div>1. Navigate to site</div> <div>2. Search for "Dell XPS 13"</div> <div>3. Select View: "Stores"</div> <div>4. Verify</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div>- Search returns same product from multiple stores</div> <div>- Each store entry shows different stock levels</div> <div>- Stock status badges correct: green (In), yellow (Low), red</div> |

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|  |  | <p>- Each store has different quantity levels</p> |  | <p>multiple stores listed showing the product</p> <p>5. Note the stock levels for each store:</p> <ul style="list-style-type: none"><li>- Store A: 15 pcs (In Stock)</li><li>- Store B: 3 pcs (Low Stock)</li><li>- Store C: 0 pcs (Out of Stock)</li></ul> <p>6. Verify stores are sorted by:</p> <ul style="list-style-type: none"><li>- Distance (nearest first)</li><li>- Then by stock availability (In &gt; Low &gt; Out)</li></ul> <p>7. Click on Store A marker on map</p> <p>8. Verify it shows "Dell XPS 13: 15 pcs"</p> <p>9. Click on Store B marker</p> <p>10. Verify it shows "Dell XPS 13: 3 pcs - Low Stock"</p> <p>11. Click on Store C marker</p> <p>12. Verify it shows "Dell XPS 13: Out of Stock"</p> <p>13. Click "Details" on Store A</p> <p>14. Verify modal shows complete inventory including other products</p> <p>15. Note Store A's distance from user (e.g., 2.5 km)</p> <p>16. Click "Details" on Store B</p> | <p>(Out)</p> <ul style="list-style-type: none"><li>- Stores sorted primarily by distance, secondarily by stock</li><li>- Map markers clickable and show correct info</li><li>- Store comparison helps customer choose nearest available stock</li><li>- Distance calculation uses Haversine formula (accurate)</li><li>- Customer can make informed decision before visiting</li></ul> |
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|         |                                                                                          |                                                                                                                                        |                                  | 17. Verify Store B's distance is greater than Store A's                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ITC-008 | Verify password hashing, validation, and security measures during registration and login | <ul style="list-style-type: none"><li>- No user exists with email "security@test.com"</li><li>- Registration page accessible</li></ul> | Password Validation and Security | <ol style="list-style-type: none"><li>1. Navigate to login-register page</li><li>2. Click "Register" in Customer section</li><li>3. Enter registration details:<ul style="list-style-type: none"><li>- Username: "security_test"</li><li>- Email: "security@test.com"</li><li>- Password: "weak" (too short)</li></ul></li><li>4. Click "Register" button</li><li>5. Verify error message: "Password must be at least 8 characters"</li><li>6. Change password to "Password123"</li><li>7. Click "Register" button</li><li>8. Verify successful registration</li><li>9. Open browser DevTools &gt; Application &gt; LocalStorage</li><li>10. Look for password in sf_user object</li><li>11. Verify password is hashed</li><li>12. Verify hash starts with "\$2y\$" or "\$2a\$" (bcrypt format)</li></ol> | <ul style="list-style-type: none"><li>- Weak passwords rejected (&lt; 8 characters)</li><li>- Valid password accepted</li><li>- Password hashed using bcrypt before storage</li><li>- Hash stored in database, not plain text</li><li>- bcrypt cost factor = 10 (as per HLD specification)</li><li>- Login validates hashed password correctly</li><li>- Wrong password rejected with generic error message</li><li>- No sensitive data exposed in console or localStorage</li><li>- Password case-sensitive</li><li>- No password visible in network requests</li></ul> |

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|  |  |  |  | <div>13. Logout</div> <div>14. Attempt login with:<ul style="list-style-type: none"><li>- Email: security@test.com</li><li>- Password: Password123</li></ul></div> <div>15. Verify successful login</div> <div>16. Logout</div> <div>17. Attempt login with:<ul style="list-style-type: none"><li>- Email: security@test.com</li><li>- Password: password123 (lowercase - wrong)</li></ul></div> <div>18. Verify error message: "Invalid email or password"</div> <div>19. Check browser console for errors</div> <div>20. Verify no plain text passwords logged</div> |  |
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StockFinder PH

A Partial Course Requirement in Apps Dev

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1st Semester 2025 - 2026

